2024 Ph H2 Q7

Section: Particles and Waves

Topic: The Standard Model

Summary of Question

A student claims that particle physics is like matryoshka dolls: look inside a particle and you will always find a smaller, similar particle.

Task: Using your knowledge of physics, comment on this statement. (3 marks)

Answer (comment)

• There are layers at first: atoms → nuclei → nucleons (protons/neutrons). But nucleons are made of quarks, while the electron is fundamental; opening it does not reveal a smaller 'similar' particle.

• In the Standard Model, quarks and leptons are treated as elementary (point-like down to current experimental limits). Hadrons are composite; leptons (e.g., electron, muon) show no substructure in deep inelastic scattering.

• Constituents are not miniature copies of the original object (quarks inside a proton are not smaller protons), and interactions are mediated by gauge bosons (e.g., photons, gluons) rather than by little solid balls.

Final comment: The analogy is only partly valid: some particles are composite, but many are fundamental with no smaller parts observed; constituents are also different in kind, not just smaller versions.

Revision Tips

For 3-mark ‘comment’ questions: make two or three distinct, correct points and finish with a clear overall judgement.

Name the layers correctly and distinguish composite (hadrons) from elementary (quarks, leptons).

It’s fine to note that future theories may go deeper, but current evidence supports quarks/leptons as fundamental.